Define Phase Workbook

Turning validated customer insights into a product concept, personas, value story, UX journey, and Hills. One file. You lead; the AI assists. Validate every AI output against your interview evidence.

READ ME FIRST (for the student)

This single file replaces the separate Define-phase input and output documents. It builds on your Discover-phase work: you are moving from "what problem exists?" to "what solution should we build, and how will we deliver it?" You lead the synthesis; the AI is a support tool, not a substitute. Simply accepting the AI's output without challenge lowers your grade.

How this works:

1. Do the PRE-WORK first (no AI): re-read your interview transcripts and HW4, find patterns, and form your own point of view.
2. Write the blue/red PRODUCT DEFINITION yourself (no AI). This anchors everything else.
3. Upload this file AND your interview transcripts/HW4 to the AI. On the first upload it will only ask for those materials — that is intentional.
4. Work the stages in order — Personas → Who/What/Wow → UX Journey → Hills. For each: the AI drafts (green), then you validate and refine with evidence (blue). Save and re-upload when the AI needs the latest full picture.
5. Finish with your Reflection — a NO-AI section you write yourself.

Color key

BLUE = you write it (validating the AI). GREEN = the AI writes it. AMBER = read it; don't edit it. A RED-edged blue box means: write it yourself with NO AI.

AI USAGE POLICY

Use AI only in the green sections and only when a step tells you to. Pre-work, the product definition, all validation, and the reflection are your own work. Evidence from customer interviews is mandatory for every claim.

Pre-Work — before you open the AI

STUDENT WRITES THIS — DO NOT USE AI HERE

Do this on your own first. You need your own point of view before the AI gives you one.

As a team, before using any AI:

- Review all customer interview transcripts and revisit HW4: Interview Customers.
- Identify recurring patterns in pain points, goals, behaviors, and decision drivers.
- Draft an initial product concept internally.

- Agree on likely customer archetypes.

Recurring patterns we found

Pain points, goals, behaviors, decision drivers:

Getting checked out by an eye doctor often feels like too much effort for the payoff you get. Hence, people like Anna and Linda wait months or years. Between exams, there are no reliable signal. Online and home tests aren't trusted, and kids memorize and game home charts quite easily. Some people avoid testing because they fear a worse result. #Ignorance is bliss.

People wanted a clear answer to whether a visit was worth it, not just a score. They also wanted to monitor more often without extra clinic visits. On the DMV and regulotory side, the goal was fewer counter failures and verified results.

Most people waited it out, adjusting screens or sitting closer instead of getting checked. Informal self-checks were crude and usually dismissed. People acted only when forced, by an expiring prescription, a school letter, or a DMV visit.

Action requires a pretty concrete trigger/event, a credible source with known accuracy, and a result pointing to a clear actionable next step. The check also had to fit into routines people already have. For the DMV, the drivers were legal compliance, identity verification, and audit-ready records.

Our initial product concept & likely archetypes

Draft concept and the customer archetypes you expect:

An at-home/anywhere vision check system, validated against clinic equipment, that tells users whether it's time to see an eye doctor or they qualify for task 'x'. It complements the exam rather than replacing it, or is a way of upkeeping the results of the exam. The test has features and elements in play for anti-cheating/anti-gamification/anti-memorization. Repeated checks show trends over time. Users can share results with their optometrist and book an appointment directly. A verified pre-screen for DMV applicants is a possible later phase, pending validation with real agencies. Even beyond that a real regulatory tool for individuals operating aircrafts and so on.

Likely archetypes:

Routine-Check Riley: Adults with glasses who notice their vision slipping but put off exams. #cantbeasked
Parent Pat: Engaged parents who want to monitor their kids' vision more often than once a year.
Licensing Supervisor Lisa: DMVs that could use verified pre-screening, easy renewals etc.

Instructions for the AI

TO THE AI ASSISTANT READING THIS FILE

This block is written for you. Follow it exactly. Do not edit it, and do not write in any blue or red-edged section. (Student: leave this section as-is.)

This is a Define-phase product workbook. You are a research assistant and brainstorming partner — the student leads the synthesis. The student will challenge your output against real interview evidence, so be grounded and never idealize.

Hard operating rules

- Treat this workbook as context, not a deliverable. Only write in the green "AI WRITES" sections; never write in blue or red-edged sections.
- Do not infer, invent, or extrapolate beyond what the student has written and the interview evidence they provide.
- Do not ask follow-up questions, except the single required request described below.
- Base every persona, story, journey, and Hill only on the student's product definition and the customer transcripts/HW4 they share. If something is not supported by that evidence, do not assert it.

GATING — your first response after reading this file

On the first upload you will see the product definition but NOT the interview materials. Do not generate personas, stories, journeys, Hills, or any analysis yet. Acknowledge and retain the product context, then give exactly this response, verbatim, and nothing else:

Please share the customer interview transcript and HW4 – Interview customers for your identified problem statement.

After the student shares the interview materials — work in stages

Then proceed one stage per turn, waiting for the student to validate each before the next:

- Stage 1 — Personas: generate 2–3 persona cards grounded in the evidence.
- Stage 2 — Who/What/Wow: pick ONE persona and write a vivid Who–What–Wow story.
- Stage 3 — UX Journey: map awareness → onboarding → 3–5 key steps → resolution for that persona, with emotions and decision points.
- Stage 4 — Hills: propose 2–4 Hills (who benefits, what capability, the wow) that together deliver the journey.

Source of truth

Later stages may arrive in a fresh chat or after the student edits the file. Always treat the most recently uploaded workbook plus the provided interview materials as your only source.

Part 1 — Product / Solution Definition

Student writes · NO AI · AI reads this as context · this anchors all later work.

STUDENT WRITES THIS — DO NOT USE AI HERE

Write this entirely yourselves, with no AI. It is the single anchor the AI uses for every later stage.

Based on your Discover-phase interview evidence, describe:

- What the product is — the concept, not a detailed feature list.
- How it addresses the root customer pain points you validated.

Avoid speculative features and "solution-first" bias — stay grounded in what customers told you.

Your product / solution definition:

Focal is a quick and standardized vision check system people can use at home or potentially anywhere. It tells them whether their vision has changed and whether it's time to see a doctor. It complements the eye exam rather than fully replacing it, filling the gap between visits where people now guess, wait, or ignore changes. It has also a place in highly regulated industries like airlines, to help get more granular on pilot's vision on a daily basis.

With Focal:

- Effort outweighs the payoff. Anna notices her prescription not being good enough months early but waits. Rebooking can be a a bit of a hassle. Focal gives her a near instant answer, at the tip of her fingers. This way a visit happens only when a result shows it's needed.
- There are now clear signal between exams. Most people judge their vision by feel, and parents can't tell whether squinting means anything. Focal also has memory and keeps a detailed record on the journey of the user.
- Self-tests are finally trustworthy. People dismissed online tests because the setup seemed like a scam/guesswork and no credible authority or source stood behind them. Kids gamed home charts.
- No fear of bad news. Some people avoid testing because they don't want to learn their vision got worse. Focal frames results as a next step.

Part 2 — Personas

2A. Team POV — Likely Personas

STUDENT WRITES THIS — DO NOT USE AI HERE
Write this entirely yourselves, with no AI.

Based on your **Discover-phase customer segments, problem definition, and interview findings**, identify **3 customer personas you believe are most likely to experience the problem and/or benefit from your product.**

For each likely persona, briefly state:

Likely Persona	Why We Believe This Persona Matters
Persona 1	*What makes this segment particularly relevant to the problem/product?*
Persona 2	*What makes this segment particularly relevant?*
Persona 3	*What makes this segment particularly relevant?*

Important: *At this point, these are **your team's hypotheses**, not proven personas. You are expressing your Product Manager point of view based on what you learned during Discover.*

Keep this section brief. **1–3 sentences per persona is sufficient.**

Your persona inputs:

Persona	**Description (***persona name & title; role; primary Job to be Done; key pain points; goals/ motivations)*
Persona 1	Routine-Check Riley (adult glasses wearer) --> This person is a screen-heavy office worker or grad student who wants to know if their vision has changed enough to see a doctor. To them, booking an exam feels like too much effort to go through for a small change, online tests aren't trusted, and younger adults often don't know where to go once they are off their parents' insurance. They want a quick and credible answer to the question of "do I need to go?"
Persona 2	Parent Pat (engaged parent) --> This is a working parent who wants to catch changes in their kids' vision between annual exams. Kids can't explain what they see, memorize home charts, and prescriptions can change within months, so they want frequent and reliable monitoring at home with records to share with their eye doctor.
Persona 3	Licensing Supervisor Lisa (DMV operations supervisor) --> This person oversees counter vision screenings and needs results that meet the state standard and hold up to audits. Failed screening cause repeat visits, results vary by staff and equipment, and records are just a checkbox. She wants fewer counter failures and verified, audit-ready results.

2B. AI-generated persona cards

AI writes · student does not edit · must be grounded in interview evidence.

THE AI WRITES THIS — paste the AI's output here; you validate it in the next section
Ask the AI for 2–3 persona cards based on your product definition and the interview evidence you shared. Paste them here.

Each card should include: persona name & title; role & context; primary Job to be Done; key pain points; goals & motivations; behaviors & technology comfort level.

AI output — persona cards:

Each card draws only on your Product Definition and the interview records in your Discovery workbook. I tag where each point comes from: **V** = virtual interview, **IP** = in-person.

Card 1: Routine-Check Riley (adult glasses wearer, screen-heavy)

Role & context: Office worker or grad student in their 20s–30s who has worn glasses or contacts for years. Riley is on screens most of the day and has no deliberate way to track changes. Supporting interviews: Riley (V), Anna Altmann (IP), Anna K. (IP), Linda (IP).

Primary Job to Be Done: Find out whether my vision has changed enough to justify an eye-doctor visit.

Key pain points

- **Effort out of proportion to payoff.** Riley's last exam took about 3 weeks to book and a long lunch. The whole visit ran about 2 hours and left them dilated for the afternoon. Riley called it "a half day of disruption just to hear that my prescription had changed 'a little.'" Anna Altmann: "it shouldn't take so much time to find out I need slightly new glasses." Linda said the cost was mostly the effort of the trip.
- **No signal between exams.** Riley goes "by feel" and has no baseline or prescription handy. Anna Altmann notices her prescription slipping 2–4 months early but "waits it out." Anna K. started sitting closer in lectures but "hadn't connected this to her vision until asked."
- **Uncertainty about where to go and what's covered.** Riley couldn't explain what insurance covered and only learned the cost at the register. Anna K. doesn't know which local doctors take her parents' plan. She said that once she's off it she'll "probably just avoid going."
- **Online tests aren't trusted.** Riley didn't believe a "within normal range" result and saw no organization behind the test. Anna K. said "okay, says who." Linda said the image-based tests she'd seen weren't accurate.
- **Fear of bad news (one interview).** Linda sometimes avoided testing for fear her prescription had increased. She went 5–6 years without a check at one point.

Goals & motivations: A quick, credible answer to "do I need to go?" Riley and Anna Altmann both want that answer without a clinic visit. Riley wants a result that says "this is different from before" or "this looks the same," not "within normal range."

What would make them act

- **Concrete triggers.** Riley went only when an expiring contacts prescription forced it. Anna K. went only when her mom booked it. Riley said "a vague feeling that I should go never works."
- **Credibility.** Riley wanted to know who stands behind the test, clear instructions (distance, glasses on or off, lighting) and some indication of how sure the result is. Riley would act if a real optometrist would "say 'okay, that's useful.'" Linda wanted medical or hospital certification and the measurement error versus hospital equipment. Anna K. would trust a result an eye doctor reviewed or friends recommended.

Behaviors & technology comfort: Riley used a free online eye test and googles symptoms. Anna K. didn't know online vision tests existed. Linda has seen vision tests on Xiaohongshu but never used a vision-testing app. Tech comfort is not otherwise evidenced.

Card 2: Parent Pat (parent monitoring kids' vision)

Role & context: Working parent of school-age children who relies on school and pediatrician screenings and a pediatric eye doctor when something seems wrong. Supporting interviews: Parent Pat (V), Fan's mother (IP). This card has the thinnest evidence of the three, with one virtual and one real interview.

Primary Job to Be Done: Know when there is a real reason to get my child's eyes checked, and catch changes early.

Key pain points

- **Behavioral signs are hard to interpret.** Pat watches for squinting, holding books close and headaches. Pat said "sitting close to a screen doesn't necessarily tell me that their eyesight has changed." Pat had also wondered about the TV habit but kept putting it off until a school letter arrived.
- **Waiting and uncertainty around follow-up.** After a school screening letter, Pat found the wait "the most difficult" part because the letter might or might not be serious. The first workable appointment was several weeks out. Insurance took a couple of calls, and Pat had to arrange time off work.
- **Home charts don't work for kids.** The real mother's children memorize the letters or symbols. Pat's younger child "treated it like a guessing game," and Pat wasn't sure about distance or chart size.
- **Annual exams are too infrequent for a high-risk family.** The real mother's family has a history of nearsightedness. She keeps all exam records and compares results over time. She said the prescription can get "significantly stronger within just six months," and she would prefer checkups about every three months. Extra visits are costly and inconvenient, and appointment waits are long.

Goals & motivations: Pat wants a reasonable reason to act: "I don't necessarily want to check their vision constantly. I want to know when there is a reasonable reason to get it checked." The real mother wants more frequent data so she can decide when to act. These two goals differ in intensity and may be separate sub-segments.

What would make them act

- **Fit and clarity.** Pat needs the test to be age-appropriate and done correctly, with a clear explanation of what the result means.
- **Professional backing.** Pat would trust it if the pediatrician or eye doctor recommended it. Pat said "one reassuring result wouldn't make me ignore a concern raised by their teacher." The real mother said it would need to be "accurate and consistent."

Behaviors & technology comfort: Pat has tried an online eye chart and is willing to try home checks. Beyond that, tech comfort is not evidenced.

Card 3: Licensing Supervisor Lisa (DMV operations supervisor)

Role & context: Operations supervisor at a state licensing office who oversees counter vision screening, staffing and the queue. Counter reps run a desktop screener and mark pass or fail. Failures get a paper form to take to an eye doctor. **Caveat:** all evidence for this card comes from two virtual interviews (Haydn's, the more detailed, and Qingyi's) and no real agency employee. Treat it as hypothesis.

Primary Job to Be Done: Confirm each applicant meets the state vision standard quickly, consistently and in a way I can defend later.

Key pain points

- **The failure path is the real cost, not the test.** In Haydn's interview the screening takes about a minute and "if you deleted the vision test... I'd save maybe four or five percent of counter time." A failure collapses the transaction, upsets the applicant and consumes a second slot later. Lisa described it as roughly two transactions plus overhead.
- **Weak evidence trail.** Lisa has a checkbox, a date and an employee ID, with no acuity number, attempt count or service record. In her words, the gap is between "we did it right" and "we can show we did it right."
- **Inconsistent administration and aging equipment.** Reps vary on retries. One unit is about 12 years old. Lisa has never seen calibration documentation. A dimming lamp was caught only by a veteran rep's instinct, and wrongly failed applicants went unreviewed.
- **Paper-heavy referrals.** Forms get lost or mis-mailed, and clearing a record is manual.

- **Integrity of any remote result.** Lisa's biggest concern is who actually took the test, since a website "doesn't lose anything" if it's wrong. She noted that a statute may require screening at the point of application.

Goals & motivations

- Fewer failed-at-counter outcomes: if "even half" arrived already cleared, she'd "notice that in my day."
- A number in the record instead of a checkbox.
- A chain she could defend to a lawyer: who took it, when, under what conditions, and who stands behind it.
- Something in it for the driver, not just the state.

Behaviors & constraints

- **Adoption.** Any change needs leadership, IT/security, legal and procurement. Lisa also said counter reps decide whether a new step actually happens, because "if it adds a step, they'll find a way around it."
- **Technology comfort:** not evidenced beyond existing screening machines.

Evidence tensions and gaps (flagged for your validation)

- **Lisa's two interviews disagree on line impact.** Qingyi's Lisa called vision screening "a bottleneck." Haydn's Lisa said it is a small part of the transaction and that document verification slows the line.
- **Memorization.** Lisa said most memorizers aren't deliberately cheating. They are often older drivers who "don't want the answer."
- **Pilots.** Your Product Definition mentions aviation and pilot use, but I found no interview evidence for it, so I did not build a persona around it.
- **Young adults without their own eye care.** Anna K. (IP), Kevin W. (IP) and Alex (V) form a possible distinct pattern: no eye doctor, insurance confusion, and low urgency. Kevin W. also said he'd only use a check folded into an existing routine like an annual physical, and a standalone app or website "wouldn't even register." This could be its own persona or a sub-segment of Riley, and it's your call.

2C. Your validation of the personas

Student writes · validate against interview evidence.

STUDENT WRITES THIS — validate the AI's output against your interview evidence

Personas must reflect real customers, not idealized users. Validate each against your interview evidence.

Students now compare, contrast and confirm *likely personas* from the Team's and AI inputs. Select 3 personas you will move forward with. For each persona: confirm the segment is real, correct any unsupported assumptions, and add insights you or the AI missed — citing the interviews that support your changes.

Your persona selections, validation, corrections, and added insights (with interview evidence):

We are going to move forward with the same three personas from our team POV: Routine Check Riley, Parent Pat, and Licensing Supervisor Lisa. The AI's cards that we just generated matched our view relatively closely, but we made a few corrections and additions based on the interviews.

Routine Check Riley (confirmed and expanded)

This persona is the one that is the best supported (evidence from Riley, Anna Altmann, Linda, and Anna K. from the interviews). We agree the main pain point is that an exam feels like too much effort for a small change. Anna Altmann notices her prescription slipping months early but waits it out. We also folded in young adults as a sub-group instead of making them a separate persona. Anna K., Kevin W., and Alex all lacked their own eye doctor, were unsure about insurance, and felt little urgency. Kevin W. added something we hadn't considered, which was that he said a standalone app wouldn't even register and he'd only do a check if it were part of something he already does (like an annual physical). As a result, Riley needs a strong trigger and a product that fits an existing routine and not just convenience.

Parent Pat (confirmed, split into two needs)

This persona is supported by Parent Pat and Fan's mother. We agree with the AI that this card has thinner evidence and that it contains two different needs. Fan's mother is highly engaged, keeping records and wanting checkups every three months because her kids' prescriptions can jump within six months. Pat only wants to know when a checkup is actually worth it. Both confirmed that kids memorize home eye charts, so any home test must change each time.

Licensing Supervisor Lisa (kept as a hypothesis)

Since all evidence comes from two virtual interviews, this persona still needs validation with a real DMV employee. The two interviews disagreed on whether the vision test slows the line. We side with Haydn's more detailed interview, which said the test itself takes about a minute and that the real cost is when someone fails and has to come back. Lisa's key needs are fewer counter failures, a real number in the record instead of a checkbox, and proof of who took the test.

We dropped pilots and airlines from our personas b/c none of our interviews supported them.

Part 3 — Who, What, Wow Story

3A. AI-generated Who, What, Wow

AI writes · student does not edit.

THE AI WRITES THIS — paste the AI's output here; you validate it in the next section

Ask the AI to pick ONE persona and write a Who–What–Wow story. Paste it here.

WHO = the selected persona (by name) · WHAT = the specific goal/job they're trying to accomplish · WOW = the vivid, memorable moment or outcome when they achieve it.

AI output — Who, What, Wow:

Persona chosen: Routine-Check Riley. This is your best-supported persona, backed by four interviews: Riley (V), Anna Altmann (IP), Anna K. (IP) and Linda (IP). The Parent Pat and Licensing Supervisor Lisa cards rest on thinner evidence, so they are weaker bases for a core value story.

WHO

Riley, an office worker in their early 30s who has worn glasses since middle school and spends nine or ten hours a day on a laptop and second monitor. Riley's last eye exam was over two years ago, and only because an expiring contacts prescription forced it. By mid-afternoon Riley's eyes are tired and sometimes a dull headache sets in. Night driving has started to feel harder: "road signs seem a bit fuzzier." Riley can't tell whether that is a real change or just a tired week, because there is "no baseline to compare against."

Riley is not careless about eye health. For Riley the exam costs more effort than it seems worth. In Riley's words, the whole experience "felt like a half day of disruption just to hear that my prescription had changed 'a little.'" Riley has tried a free online test and didn't believe it: no one recognizable stood behind it, and "within normal range" didn't answer the real question.

WHAT

The job: Know, quickly and credibly, whether my vision has actually changed and whether it is worth seeing an eye doctor.

In the interviews, the answer has to meet these conditions:

- **It must be quick.** Riley and Anna Altmann both want an answer without a clinic visit.

- **It must be credible.** Riley wants to know "who's behind it," instructions that make it clear the test was done right, and an indication of how sure the result is. Linda wants hospital or medical certification and the measurement error versus hospital equipment.
- **It must give a next step, not a score.** Riley wants "this is different from before, you should get it looked at" or "this looks the same, you're fine for now."
- **It must be usable by a real doctor.** Riley said they would act if an optometrist would say "okay, that's useful."

WOW

Illustrative scenario built from interview evidence, not a prediction of exact product behavior.

It's a Tuesday night. Riley has had a headache behind the eyes all week and just squinted at a road sign on the drive home. Two years ago Riley would have shrugged and kept reordering the same contacts. Tonight, Riley does a short Focal check at the kitchen table, following clear on-screen instructions about distance, whether to keep glasses on and lighting, so there is no guessing whether it was done right.

Within minutes Riley gets a plain answer: **this is different from your last check, and it's worth getting looked at.** There's no confusing score and no "within normal range." The result states how it compares to Riley's earlier checks and what it was validated against. Focal frames it as a next step, not a verdict.

Riley can send the result to an optometrist and book an appointment from the same place. For the first time, Riley isn't guessing whether a visit is worth three weeks of waiting and a long lunch. The visit happens because the result showed it was needed.

If Riley's result had instead said "this looks the same as before, you're fine for now," Riley would have had a trusted reason to skip the visit and stop worrying.

The wow is the moment of certainty: after years of going by feel, Riley gets a credible, comparable answer in minutes, from the kitchen table.

Evidence notes for your validation

- **Not covered by this story:** Fear of bad news (Linda) and the young-adult sub-group with no eye doctor (Anna K., Kevin W., Alex) are only lightly reflected. The result framing speaks to fear. The scenario does not address a user who has no provider to send the result to.
- **Routine fit:** Kevin W. said a standalone app "wouldn't even register" and a check would need to fit an existing routine. This story assumes Riley starts the check on their own, driven by a concrete trigger (night driving, headaches). Riley said "a vague feeling... never works," so a trigger is realistic for Riley but not for Kevin.

- **Claims not backed by interviews:** The workbook doesn't establish how Focal's validation or accuracy information would be displayed. I only described it as visible, based on what Riley and Linda asked for.

3B. Your refined Who, What, Wow

Student writes · this becomes your core value narrative.

STUDENT WRITES THIS — validate the AI's output against your interview evidence
Refine the story so it is accurate and compelling. This becomes your core value narrative.

Check: does the WHO accurately represent the persona from interviews? Is the WHAT specific and grounded in real customer goals? Is the WOW differentiated and believable? Then state what you changed to strengthen it.

Your refined Who, What, Wow (and what you changed):

WHO: Riley, an office worker in their early 30s who has worn glasses since middle school and spends most of the day on screens. Riley's last eye exam was over two years ago and only because an expiring contacts prescription forced it. Lately Riley has had tired eyes, headaches, and fuzzier road signs at night but can't tell if it's a real change or just a long week.

WHAT: Riley wants to know quickly and credibly whether their vision has actually changed and whether it's worth seeing an eye doctor without having to give up half a day to find out.

WOW: After squinting at a road sign on the drive home, Riley does a short Focal check at the kitchen table, which has clear instructions so there's no guessing whether it was done right. Within minutes, Riley gets a plain answer saying that this is different from their last check and that it's worth getting looked at. Focal shows Riley nearby eye doctors that take their insurance and lets them share the result and book an appointment in one place. For the first time, Riley isn't guessing. If the result had said that this looks the same and she's fine for now, Riley would have had a trusted reason to skip the visit and stop worrying.

What we changed:

- Shortened the story to focus on the key moment instead of every interview detail.
- Added help finding an in-network eye doctor. The AI's version assumed Riley already had a provider to send results to. Anna K. didn't know which local doctors took her insurance, and Riley was confused about coverage, so the wow had to include that step.
- Kept the result framed as a next step rather than a score. This matters for people like Linda, who avoided testing for fear of bad news.

- Kept the concrete trigger (headaches, night driving). Riley said that a vague feeling that she should go never works, so the story needed a real reason for Riley to start the check.

Part 4 — User Experience Journey

4A. AI-generated UX journey

AI writes · student does not edit.

THE AI WRITES THIS — paste the AI's output here; you validate it in the next section

Ask the AI to map a simple UX journey for the chosen persona, from awareness to value. Paste it here.

Include: Awareness (how they first learn of the product) · Onboarding (first encounter / getting started) · 3–5 key interaction steps · Resolution (achieving the goal and the WOW) — with brief descriptions, user emotions, and decision points at each step.

AI output — UX journey:

Stage	What Riley does	What Riley feels	Decision point	Evidence
1. Awareness / trigger	Notices tired eyes, a dull headache behind the eyes, or fuzzier road signs at night. Googles symptoms, asks a coworker who wears glasses, or jokes with friends. Hears about Focal this way.	Unsure whether it's "real or just a tired week." Mildly worried but not urgent.	Ignore it again, or check now?	Riley (V): "a vague feeling that I should go never works." Riley acts on a hard deadline or something that scares them, like a close call while driving. Riley (V): googles symptoms, asks a coworker.
2. Onboarding / trust check	Opens Focal and decides whether to believe it. Looks for who stands behind it, how it was validated, and how to set it up (distance, glasses on or off, lighting).	Skeptical, "okay, says who." Fears being told something is wrong because of a setup mistake.	Trust this enough to take it, or close it like the free online test?	Riley (V) wanted "who's behind it," clear instructions and some idea of "how sure it is." Anna K. (IP): "okay, says who." Linda (IP): wanted medical or hospital certification and the error versus hospital equipment.
3. Key step A: take the check	Follows on-screen instructions and does a short check at home, in about the time of a coffee break.	Focused, a little anxious. Possible dread of a worse result.	Redo it if unsure it was done right, or accept the result.	Riley (V): liked a roughly five-minute online test. Alex (V): did one test twice and got different results, which "made me trust it even less." Linda (IP): sometimes avoided testing for fear her prescription had increased.
4. Key step B: see the result	Gets a plain answer comparing this check to Riley's earlier one, with a confidence indication and a next step, not a bare score.	Relief or concern, but clarity either way.	"Different, get it looked at" vs. "same, you're fine for now."	Riley (V): "within normal range" tells me nothing. Wants "this is different from before, you should get it looked at" or "this looks the same, you're fine for now."
5. Key step C: act on it	If different: sees in-network eye doctors, shares the result with one and books, all in one place.	Hopeful, a little impatient about the wait.	Book now, or put it off ("next month," then two years)?	Anna K. (IP): doesn't know which doctors take her insurance. Riley (V): couldn't explain what insurance covered. Riley (V): would act if an optometrist would say "okay, that's useful." Riley (V): "I tell myself I'll go next month, and then it's been two years."
6. Resolution	If different: attends a visit for a reason the result supports. If same: skips the visit with a trusted reason and stops worrying. Later repeats the check when something feels off, building a history over time.	Confident, no longer guessing.	Whether to use Focal again before the next exam.	Product Definition: repeated checks show trends over time. Anna Altmann (IP) wants a quick test "when she feels her prescription starting to slip."

Emotional arc

Uncertain and mildly worried → skeptical → wary but willing → anxious during the check → relieved by clarity → (if different) impatient or tempted to delay → confident.

Friction points and risks, from the evidence

- **The comparison needs history.** The result in your WOW says "different from your last check." In Riley's interview there is "no baseline to compare against" and the prescription is "in an email somewhere, or on a card in a drawer." The interviews don't show how a first-time user gets a comparison, so the first-use experience is an open design question.
- **Trust is decided early.** Riley rejected the online test because no one recognizable stood behind it and because of a "buy glasses" feel. Alex (V) also read a "buy glasses" button as a sales funnel.
- **Booking can stall after a clear result.** Alex (V) said a "probably fail" result would lead to sitting on it for a while. Riley (V) says the same about putting things off. Linda (IP) avoided testing for fear of worse results.
- **Routine fit is unproven.** Kevin W. (IP) said a standalone app "wouldn't even register," and the journey assumes Riley starts the check from a trigger. That assumption fits Riley (V) but not Kevin.

Evidence gaps (not asserted in the journey)

- **How Riley first hears of Focal** is not covered by the interviews. The awareness step uses only how Riley currently handles uncertainty (googling, asking a coworker, friend recommendations for Anna K.).
- **Cost and willingness to pay** came up only as something Riley said the interviewer should have asked (V). I did not include any pricing step.
- **Following up after the visit** is not covered by any interview, so I did not extend the journey beyond the visit.

4B. Your evaluation of the UX journey

Student writes.

> **STUDENT WRITES THIS — validate the AI's output against your interview evidence**
> Evaluate the journey against how your persona really behaves.

Consider: are critical steps missing? Are any steps unnecessary or confusing? Does it realistically reflect how your persona would interact? Are there friction points to smooth out? Does it lead logically to the WOW?

Your evaluation:

Overall, the AI's journey matches how Riley really behaves. It starts with a concrete trigger like headaches or fuzzy road signs, includes a trust check before Riley commits, and ends with a clear next step instead of a score, which is what Riley asked for in the interview. It also leads logically to our WOW.

We did find a few gaps.

We are missing a starting baseline --> The journey and our WOW both say "different from your last check," but a new user doesn't have an earlier check to compare against. Riley said there's no baseline and their old prescription is in an email somewhere. The first check needs to work as a baseline, ideally with a way to enter an old prescription so the first result still means something.

Awareness is weak --> The interviews don't show how Riley would find Focal. The msot realistic channels from the evidence are work of mouth (Riley asks a coworker / Anna K said she'd trust it if friends recommended it) and an eye doctor recommending it, which would also solve the trust in step 2.

Booking can stall --> Even with a result that clearly says to get it looked at, Riley said that she tells herself she'll go next month and then all of a sudden it's been two years. Alex said he'd sit on it for a while. The journey should include a follow-up reminder if Riley doesn't book.

Consistency matters --> Alex got different results when he took an online test twice and stopped trusting it. If Riley's setup looks wrong (distance, lighting, etc), Focal should tell them before showing a result instead of giving a bad answer.

Routine fit was not covered. Kevin W said a standalone app wouldn't even register. This journey works for Riley because Riley has a trigger, but people without a trigger may need Focal built into something that they already do like an annual physical. We see this as a later question and not a change to Riley's journey.

4C. Your refined UX journey (optional)

Student writes · optional · you may use design tools and embed/link visuals.

STUDENT WRITES THIS — validate the AI's output against your interview evidence
If significant changes are needed, build your refined journey here. You may modify the AI version or create a new one, and may embed or link Figma/Miro/other visuals. Explain what you changed and why.

Your refined UX journey (and what you changed and why):

Part 5 — Product Delivery Hills

5A. AI-generated Hills

AI writes · student does not edit.

THE AI WRITES THIS — paste the AI's output here; you validate it in the next section

Ask the AI to propose 2–4 Hills based on the UX journey. Paste them here.

Each Hill: name/title · who benefits (which persona) · what they can do (capability delivered) · the wow (meaningful outcome). The Hills should collectively deliver the complete journey.

AI output — Hill definitions:

Hill 1 — Start a check with confidence

- **Who:** Riley, an adult glasses wearer who notices possible vision changes but distrusts online tests.
- **What:** Understand what Focal can and cannot tell them, review its validation information, and follow clear setup instructions.
- **Wow:** Riley can start a check knowing what supports the results and how to use the test correctly, without guessing about distance, lighting, or glasses.

Hill 2 — Get a clear result and next step

- **Who:** Riley, who wants to know whether a change is worth seeing an eye doctor about.
- **What:** Complete a short vision check and receive an understandable result, including uncertainty and a suggested next step.
- **Wow:** Riley gets useful guidance within minutes instead of a score that leaves them unsure what to do.

Hill 3 — Turn a result into an appointment

- **Who:** Riley, who often delays exams because finding a doctor and understanding insurance feels difficult.
- **What:** Find nearby eye doctors who accept their insurance, share the check result, and access appointment booking.
- **Wow:** Riley can move from “I should get this checked” to arranging a visit without having to start a separate search.

Hill 4 — Understand changes over time

- **Who:** Riley, who currently relies on memory and how their vision feels between exams.
- **What:** Save a first check as a baseline and compare later checks completed under consistent conditions.
- **Wow:** Riley can see a clear history of results instead of guessing whether their vision has changed.

5B. Your evaluation & refined Hills

Student writes · these bridge Define and Develop.

STUDENT WRITES THIS — validate the AI's output against your interview evidence
Refine the Hills so they represent all the work and align with your vision.

→ Bridges to the Develop phase

Evaluate: do these Hills collectively represent all the work needed? Are any missing or redundant? Are the Who/What/Wow components clear for each? Do they align with your product vision and journey? Note that in the Develop phase you will write epics for each Hill, then propose PM coding assignments to wireframe or prototype those epics.

Your refined Hills (and your evaluation notes):

Our evaluation

The four Hills cover the main parts of Riley's journey: trusting the test, taking it, understanding the result, and getting help when needed. They also cover future checks, which support our goal of helping people monitor their vision between eye exams.

We kept all four because each solves a different problem. Hill 2 explains the result of today's check, while Hill 4 helps Riley understand changes over time.

We made three changes. First, we made the first-time experience clearer. Focal cannot say that Riley's vision has changed if there is no earlier result to compare it with. The first check should create a starting point for future checks.

Second, we removed the idea that an unchanged result means Riley is "fine" or can safely skip an eye exam. Focal should explain the limits of the check because it does not replace a full exam.

Third, we made the booking Hill more realistic. Finding doctors, checking insurance, and booking appointments are useful goals, but we still need to confirm how these connections would work.

Our refined Hills

Hill 1 — Know what to expect

Riley can understand what Focal checks, what supports its results, and how to set it up correctly, so they could complete a valid first check and establish a starting point for future comparisons.

Hill 2 — Know what to do next

Riley can finish a short check and get a clear, non-alarming explanation of the result and the next step. If the result is uncertain, Focal explains that instead of giving a plausible answer.

Hill 3 — Get help with less effort

When a professional medical service is recommended, Riley can have a clear care plan without having to restart the entire process on their own. Focal helps them find doctor available, check insurance information, and determine how to schedule an appointment or contact the clinic.

Hill 4 — See changes over time

Riley can save the first result and compare future checks with it, so they have a record instead of relying only on memory. Focal clearly explains when results cannot be compared reliably.

These Hills match our vision of supporting people between eye exams. For the Develop phase, we need to work out how to guide test setup, handle unclear results, compare checks fairly, and connect users with clinics. We also need to test whether Riley understands and trusts the guidance enough to act on it.

Part 6 — Reflection & Key Learnings (NO AI)

STUDENT WRITES THIS — DO NOT USE AI HERE
Write this entirely yourselves. This is where you demonstrate professional judgment — the core of your grade.

Address each point, being specific and referencing concrete examples from your work:

- The most valuable insights from developing your personas and user journey.
- What the AI helped you see or structure that you might have missed.
- Where the AI missed the mark or misunderstood your customer insights.
- How your product vision evolved from the Discover phase to the Define phase.

- Which assumptions were validated or invalidated.
- Your biggest remaining uncertainties about the personas or UX.
- How this work will inform your Develop-phase next steps.

Your reflection & key learnings:

Developing personas and a user journey showed us that our biggest challenge isn't building an accurate vision test. It is getting people to trust it and act on it. In our interviews, people like Riley, Anna Altmann, and Linda already knew that their vision might be slipping, yet they still waited months and in some cases years. Things like effort, fear of bad news, and low urgency mattered more than access to a test. Mapping Riley's journey made it clear that every step from deciding to trust Focal to actually booking a visit is a place where users can drop off.

We found that the AI was most helpful in organizing our evidence. It tagged each persona point with the interview it came from, which showed us that Riley was well supported and that Parent Pat and Licensing Supervisor Lisa had less evidence. It also pointed out things that we had overlooked, such as the two Lisa interviews disagreeing about whether vision screening slows the DMV line and the fact that a first-time user has no earlier result to compare against.

At the same time, we found that AI missed the mark in some other places. Its Who/What/Wow assumed Riley already had an eye doctor to send results to, even though Anna K. didn't know which local doctors took her insurance. Its journey also ended once Riley got a result, even though Riley and Alex both said they would put off booking afterward. We added help finding an in-network doctor and a follow-up reminder to fix this.

Our product vision changed from Discover to Define. We started with a broad idea that included DMVs and even pilots. Through the personas and journey, we narrowed our focus to individuals like Riley, who need a credible answer to "do I need to go?" We dropped pilots because none of our interviews supported them, and we kept the DMV as a later possibility that still needs validation with a real agency employee.

Our interviews confirmed that exams feel like too much effort for small changes, that people don't track their vision between exams, and that online tests aren't trusted without a credible source behind them. They challenged our assumption that convenience alone would get people to act. Kevin W. said a standalone app "wouldn't even register," and Linda avoided testing out of fear. Fan's mother also showed that some parents already track results closely and want more frequent checks rather than better record-keeping.

Our biggest remaining uncertainties are 1. whether Riley would trust Focal enough to act on its result, 2. how the first check can be useful without an earlier result, and 3. whether people without a clear trigger would ever start a check. The Lisa persona also relies only on virtual interviews.

In the Develop phase, we are going to write epics for each of our four Hills, focusing first on guiding test setup, explaining results clearly, and creating a starting baseline. We also want to

test early wireframes with real users like Riley to see whether they understand and trust the guidance enough to book a visit.

Submission & Checklist

Submit one package per team: this completed workbook plus any interview transcripts / HW4 materials you provided to the AI.

Before submitting, confirm:

- ☐ Pre-work done before any AI use (patterns, initial concept, archetypes)
- ☐ Product definition written with NO AI (Part 1)
- ☐ AI personas pasted and validated against interview evidence (Part 2)
- ☐ Who/What/Wow drafted by AI and refined into your value narrative (Part 3)
- ☐ AI UX journey evaluated, with a refined version if needed (Part 4)
- ☐ AI Hills evaluated and refined; Who/What/Wow clear for each (Part 5)
- ☐ Reflection written with NO AI (Part 6)
- ☐ Every persona, journey, and Hill claim backed by interview evidence
- ☐ Interview transcripts / HW4 included in the submission

Appendix — Access Rules

Section access legend

Box style	Meaning
Blue	Student writes (validating the AI); AI may read as context but must not edit.
Green	AI writes; you then validate and refine in the next section.
Amber	Read it; do not edit it.
Blue with red edge	Student writes with NO AI — your own work only.

Read / write enforcement the AI follows

- AI READ: YES → the AI may use the content as context.
- AI WRITE: YES → the AI may generate content there (green sections).
- AI READ: NO → the content is kept out of AI context.
- AI WRITE: NO → the AI must not generate content there.
- Any section not explicitly marked is treated as READ: NO / WRITE: NO.